INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

BUSINESS CONTINUITY AND DISASTER RECOVERY ASSIGNMENT

GRC205 – Incident Management and Business Continuity | Week 21

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EXECUTIVE SUMMARY

This report reviews Northbridge Skills Institute's needs for business continuity and disaster recovery after a flood and power failure shut down its main site and local systems. It also looks at the five business functions in the assignment: how the disruption affects learners and staff, recovery goals, backup plans, strategy costs, and the steps needed to restore services.

The analysis shows that teaching and assessment are the top priorities because of the scheduled exam at 11:00 on Day 0. Still, technical recovery must start with the network, AUTH01, security checks, and DB01 before other applications can be restored. The estimated cost of disruption is NGN 1,200,000 for eight hours and NGN 3,360,000 for 24 hours if no recovery steps are taken. These numbers are estimates for the exercise, not actual losses.

The recommended option is the warm recovery package, which includes independent protected backups and training/testing for N 2,100,000 per year. This fits the N3,000,000 budget, but the recovery schedule does not restore assessment service by 11:00. So, temporary assessment arrangements are needed. The plan also suggests making backups more independent, testing restorations, clarifying decision-making, and setting clear improvement actions.

TASK 1: BUSINESS IMPACT ANALYSIS

1.1 Context and assumptions

At 08:00 WAT on Monday, Day 0, flooding and power failure make the main site and local systems unavailable. Facilities estimates that access will be restricted for 72 hours. Microsoft 365 identity and email remain available through mobile internet. The alternate facility can provide 25 workstations from 10:00, while 40 of the 120 learners sitting an 11:00 examination require centre workstations. Two technicians are available for disaster recovery.

The BIA identifies the effect of disruption on each function and the minimum service that can be delivered safely. Hourly costs are treated as separate and non-overlapping. The 8-hour and 24-hour calculations assume no recovery or continuity action. We add the one-off examination rescheduling cost because no acceptable assessment service is available by 11:00.

1.2 Five-function BIA

Function / owner	Users and resources	Impact / dependencies	Minimum service / priority
F1 Teaching and assessment; Academic Director / Examinations Lead	120 learners; LMS01, DB01, network, AUTH01, invigilators, candidate list and secure assessment materials.	Exam starts 11:00. Learner welfare, fairness, assessment integrity and reputation at risk. NGN 60,000/hour plus NGN 120,000 once if rescheduled.	Priority 1. Use a validated assessment service for 80 own-device learners and a safe, controlled arrangement for 40 centre-dependent learners. Do not exceed 25 workstation capacity; use approved sittings or defer if fairness and integrity cannot be maintained.
F2 Fees and finance; Finance Manager	Learners, finance staff, payment provider, FIN01, DB01 and transaction records.	Fee confirmation due 16:00 Day 0. Provider is available but local receipts/wallet records cannot be updated. NGN 40,000/hour.	Priority 2. Use restricted temporary transaction register, dual approval and provider reconciliation before final confirmation.
F3 Admissions and registration; Admissions Manager	Applicants, admissions staff, REG01, DB01, applications and change logs.	Admissions close 18:00 Day 0. Local entry unavailable. NGN 20,000/hour.	Priority 3. Accept time-stamped applications through approved secure channel, acknowledge receipt and use second-person review before entry.
F4 Staff payroll; HR / Payroll Manager	90 staff, HR01, DB01, payroll team, bank submission channel and approval records.	Bank deadline 12:00 Day 1. Incorrect or duplicate instructions can affect pay. NGN 10,000/hour.	Priority 4. Preserve approved payroll file, log changes, require maker-checker approval and validate bank file before submission.
F5 Records and certificates;	Learners/graduates, records staff, FS01,	48-hour internal service commitment;	Priority 5, with urgent cases escalated. Log requests, verify

Registrar / Records Manager	network, AUTH01 and protected files.	integrity and access restrictions are essential. NGN 5,000/hour.	identity/authority and release files only through approved controls.

1.3 Disruption cost calculations

The supplied hourly rates total NGN 135,000 per hour. The examination rescheduling charge is included once in both scenarios because the scenario assumes the 11:00 service is missed.

Function	Hourly rate	8-hour impact	24-hour impact
F1 Teaching and assessment	NGN 60,000	8 × 60,000 + 120,000 = NGN 600,000	24 × 60,000 + 120,000 = NGN 1,560,000
F2 Fees and finance	NGN 40,000	8 × 40,000 = NGN 320,000	24 × 40,000 = NGN 960,000
F3 Admissions and registration	NGN 20,000	8 × 20,000 = NGN 160,000	24 × 20,000 = NGN 480,000
F4 Staff payroll	NGN 10,000	8 × 10,000 = NGN 80,000	24 × 10,000 = NGN 240,000
F5 Records and certificates	NGN 5,000	8 × 5,000 = NGN 40,000	24 × 5,000 = NGN 120,000
TOTAL	NGN 135,000/hour	NGN 1,200,000	NGN 3,360,000

There are two main limitations. First, the rates are based on assumptions and might not match real lost revenue, overtime, supplier costs, or actual recovery time. Second, the calculation does not put a financial value on learner welfare, missed chances, reputation damage, fairness in assessment, or corrupted records. So, financial impact is only one part of the decision.

Business priority is different from the technical order of restoration. F1 is the top business priority because of the 11:00 exam. But LMS01 cannot be restored before the network, AUTH01, security checks, and DB01. Risks include AUTH01 failure blocking several applications, DB01 corruption affecting four functions, an incomplete backup chain making the latest checkpoint unusable, only 25 workstations for 40 learners, and production and backup administration sharing the same administrator account.

TASK 2: RECOVERY OBJECTIVES AND BACKUP DESIGN

Recovery Time Objective (RTO) is the target time within which a function should be restored after disruption. Recovery Point Objective (RPO) is the maximum acceptable data-loss period measured backwards from disruption. Maximum Tolerable Downtime (MTD) is the longest period a function can be unavailable before the impact becomes unacceptable. These are proposed planning targets and must be confirmed with the function owners.

Function	RTO	RPO	MTD and justification
F1 Teaching and assessment	3 hours for minimum acceptable service	1 hour	4 hours; exam deadline makes an earlier continuity decision necessary.
F2 Fees and finance	4 hours for controlled fee capture	1 hour	8 hours; fee confirmation is due at 16:00.
F3 Admissions and registration	6 hours for secure intake	1 hour	10 hours; admissions close at 18:00.
F4 Staff payroll	12 hours for validated processing	1 hour	20 hours; bank submission is due at 12:00 Day 1.
F5 Records and certificates	24 hours for priority requests	8 hours	48 hours, matching the internal service commitment.

A service is not recovered just because a server is running. The application or file service must pass technical checks, a business owner must validate a sample transaction, access restrictions must be checked, data must be reconciled, and acceptance must be recorded. There should be a buffer between RTO and MTD to allow time to fix any problems.

2.2 Checkpoint ages and RPO gaps

Checkpoint	Latest checkpoint	Age at 08:00	Assessment
DB01 B01 full + B02 incremental chain	07:00 Day 0	1 hour	Possibly one hour data-loss window; meets 1-hour RPO only if the full chain passes integrity and dependency checks.
AUTH01 B03 snapshot	00:00 Day 0	8 hours	Exceeds proposed 1-hour RPO. Monitor after midnight, but will require validation and controlled reapplication.
FS01 B04 full backup	23:00 Day -1	9 hours	Exceeds proposed 8-hour RPO by one hour. New/changed files require evidence-based reconciliation.

Disruption reference time is 08:00. The restoration of DB01 does not in itself guarantee that every payment or registration up to 08:00 will be included. Provider transaction references and the approved temporary logs will be of assistance in the reconciliation process, but you can't assume that a record is recoverable without some form of proof. Any changes to the AUTH01 and FS01 files after their respective checkpoints will need to be recreated. The full backup from Day -3 is another recovery point, although it is older and should be checked prior to use.

2.3 Backup and restoration design

For the DB01, it should be backed up every 15 mins (logs), daily, and weekly (full). Restoration should layer the full backup, daily backups, and logs in order.

For AUTH01: Backs up at least hourly and whenever major changes occur.

For FS01: Backs up hourly (incremental) and daily (full).

- Apply 3-2-1: maintain at least three copies, on two storage types, with one offsite. At least one copy should be offline or immutable and independently administered.
- Encrypt data both when it is moving and when it is stored. Use role-based access, multi-factor authentication, separate privileged accounts, and require two people to approve backup deletion or changes to retention.
- Keep production administration and backup administration separate. No single account should be able to change production and delete all recovery copies without being noticed.
- Document daily, weekly, and monthly retention periods, following record and privacy rules. Keep a full backup that is not connected to the network and rotate protected copies regularly.ation, malware/security checks where appropriate, database coherence checks, access-control checks and business-owner transaction tests. Record duration, evidence and corrective actions.

Having offsite storage does not guarantee you can recover data. The provider's independent-access controls are not documented, and encrypted backups may be useless if keys or credentials are lost. A live replica can also copy accidental deletions, corruption, or malicious changes. You won't know these points are covered until you test independent access, key recovery, retention protection, and full restoration.

TASK 3: RECOVERY STRATEGY AND BUDGET

Option	Base annual cost	Required additions	Total	Assessment
Cold	NGN 600,000	NGN 400,000 backups + NGN 200,000 training/tests	NGN 1,200,000	Within budget but 24–48-hour recovery and up to 24-hour data loss do not meet urgent needs.
Warm	NGN 1,500,000	NGN 400,000 backups + NGN 200,000 training/tests	NGN 2,100,000	Within budget; 3–5-hour first service and 1–6-hour data-loss window, but may miss the exam and 1-hour RPO.
Hot	NGN 2,400,000	NGN 400,000 backups + NGN 200,000 training/tests	NGN 3,000,000	Fits exactly; about 1-hour first service and up to 15-minute data loss, but leaves no budget headroom.

I recommend the warm package with both additions for NGN 2,100,000 per year. This leaves NGN 900,000 under the limit and covers independent protected backups and training/testing.

However, the 3–5-hour estimate does not guarantee service by 11:00. The modelled drill restores LMS01 at 11:45, so continuity plans are still needed. If the business owner needs recovery in about an hour, leadership should look at the hot package and check supplier claims before approving.

The hot package has the strongest stated recovery. The hot package offers the fastest recovery but uses the whole budget and still relies on staff, supplier performance, and backup quality. The cold option is cheaper but means more downtime and risk of data loss. Remaining risks include only having two technicians, limited workstations, unclear backup access independence, and possible data corruption. The executive sponsor and finance authority should formally accept any risks that are higher than the organization's comfort level.0 24-hour estimates support the business case but cannot establish annual ROI on their own. Approval also requires outage frequency, likely duration, incident history, supplier evidence, implementation costs, test results and non-financial impact assessment.

TASK 4

The plan covers F1–F5, shared infrastructure, and the 72-hour closure. The Institute Director is the main decision-maker, with the Deputy Director as backup. The plan activates if the site or a key shared system is down for more than 30 minutes, a deadline is at risk, or Facilities says the site is unsafe. The BC/DR Coordinator keeps the situation log, records decisions, escalates unresolved risks, and gives hourly updates during recovery.

The continuity team handles minimum services and arrangements for learners and staff. The DR team restores infrastructure and applications. Facilities checks if the site is safe. Incident Response gets involved if there are signs of unauthorized access, suspicious admin activity, backup tampering, malware, or unexplained data changes. Flooding by itself does not mean a cyberattack, but keep logs if a compromise seems possible.

4.2 RACI matrix

Activity	Accountable	Responsible	Consulted / informed
Activate plan	Institute Director	BC/DR Coordinator	Facilities, IT Lead, function owners
Temporary assessment	Academic Director	Examinations Lead and Learner Support	IT, accessibility support, invigilators
Validate backups	IT Lead	DR technicians	Incident Response and data owners
Restore systems	IT Lead	DR technicians	Function owners, BC/DR Coordinator
Stakeholder communication	BC/DR Coordinator	Communications / Learner Support	Director and function owners
Reconcile outage records	Relevant function owner	Named processing staff	IT, Finance / internal control

Approve return to primary site	Institute Director	Facilities Manager and BC/DR Coordinator	IT, security, function owners

4.3 Temporary procedures

F1 assessment: By 09:00, confirm the candidate list, accessibility needs, devices, and secure method. The 25-workstation site cannot seat all 40 learners who need centre workstations at once. Do not go over safe capacity. If academic leadership approves and it fits assessment rules, use controlled sittings or authorized rescheduling with identity checks, invigilation, and secure question handling. The 80 learners using their own devices should only take the online assessment if their identity, availability, and integrity are confirmed.

F2 finance: record provider transaction reference, amount, time, learner identity, preparer and approver in a restricted register. Each should be reconciled against provider reports and restored FIN01/DB01.

F3 admissions: use a controlled intake form or approved secure mailbox. Each should have a timestamp, same acknowledged, restrict access and record later entry into REG01.

F4 payroll: preserve the last approved file and change register. Also advise that only two authorised people must validate employee changes, account details, totals and bank-file integrity. Finally, retain approval and submission evidence before the Day 1 deadline.

F5 records: accept requests through a controlled channel, verify identity and authority, and prioritise requests approaching the 48-hour commitment. Do not distribute files from personal devices or uncontrolled storage. Log each release and preserve access restrictions. One person must reconcile temporary records, another must check them, and they should close them only after comparing them with restored systems.

4.4 DR runbook

Order / owner	Action and prerequisite	Completion check / contingency
1. BC/DR Coordinator	Activate plan; establish log, scope and decision owners.	Record decision and time. Escalate unresolved authority issues to Director.
2. Facilities Manager	Confirm alternate site safety, power, connectivity, occupancy and accessibility.	If unsafe or insufficient, use approved remote service or defer; never exceed capacity.
3. DR Technician 1	Restore recovery network; no prerequisite.	Test connectivity and segmentation; pause if checks fail.
4. DR Technician 1	Restore AUTH01 after network; validate snapshot and privileged access.	Check authentication, groups and logs; use logged break-glass access only if approved.
5. IT / Security Lead	Run security and backup checks after AUTH01.	Verify chain, integrity and access independence. Quarantine suspect backups and engage Incident Response.

6. DR Technician 1	Restore DB01 after security checks.	Run consistency and representative-record tests. Stop dependent apps if checks fail.
7. DR technicians	Restore LMS01, FIN01, REG01 and HR01 after DB01; restore FS01 after AUTH01/security checks.	Function owner tests service; roll back to known-good state or continue approved manual process if failure.
8. Function owners	Reconcile temporary records and approve service resumption.	Sign reconciliation and exceptions; hold unresolved transactions.
9. Director / Facilities	Approve return after safety clearance and business validation.	Record clearance, controlled data transfer, access checks and final sign-off.

4.5 Communication matrix and learner notice

Audience	Channel / timing	Owner / message
Learners and applicants	Official email, portal notice if available, approved SMS; initial notice and updates	Communications / Learner Support: service impact, exam arrangements, accessibility and next update.
Staff and function owners	Email and approved staff channel; hourly during active recovery	BC/DR Coordinator: roles, manual procedures, approved records and escalation.
Payment provider / bank	Verified contact route as needed	Finance Manager: transaction references, reconciliation and payroll deadlines.
Leadership / relevant authorities	Situation reports and formal notice when required	Director / Compliance: impact, decisions, risks and notification assessment.

Initial learner notice

Dear Learners,

Please be informed that flooding and a power failure have made the Institute’s main site and some local systems temporarily unavailable. Facilities currently estimates that the site may remain inaccessible for up to 72 hours. At this stage, no cyberattack has been confirmed.

The Institute is activating its continuity arrangements and working to uphold key services. Learners scheduled for the 11:00 examination should check their official email and approved Institute channels for instructions on the assessment arrangements. If you require an accessibility adjustment or assistance, contact Learner Support through the official channel.

We shall provide a notice as soon as arrangements are confirmed and communicate any changes directly. Please avoid sharing unverified information or sending personal or payment details by unofficial channels. We apologise for any inconvenience and reassure them that we shall prioritise learner safety, fair assessment, and the protection of records.

Thank you

Northbridge Skills Institute

TASK 5: RECOVERY SCHEDULE AND EXERCISE

5.1 Modelled baseline schedule

The schedule uses the specified warm-site times. Each task requires one technician for the full duration, and tasks cannot be split. The two-technician limit is followed. This is a modelled baseline, not a real test.

Action	Technician	Start	Finish	Prerequisite
Recovery network (30 min)	Tech 1	8:00 AM	8:30 AM	None
AUTH01 restoration (30 min)	Tech 1	8:30 AM	9:00 AM	Network
Security and backup checks (30 min)	Tech 1	9:00 AM	9:30 AM	AUTH01
DB01 restoration (90 min)	Tech 1	9:30 AM	11:00 AM	Security and backup checks
FS01 restoration (60 min)	Tech 2	9:30 AM	10:30 AM	AUTH01 and security checks
LMS01 restoration (45 min)	Tech 1	11:00 AM	11:45 AM	DB01
FIN01 restoration (30 min)	Tech 2	11:00 AM	11:30 AM	DB01
REG01 restoration (30 min)	Tech 2	11:30 AM	12:00 PM	DB01
HR01 restoration (45 min)	Tech 1	11:45 AM	12:30 PM	DB01

The assessment service is available at the earliest by 11:45, after LMS01 is restored and checked, which is 45 minutes after the scheduled exam start. All five functions are restored by 12:30. The recovery order is network, AUTH01, security and backup checks, DB01, then LMS01. The exam deadline needs an approved workaround nor deferral, even if technical recovery goes as planned.

Function	Elapsed recovery	RTO result	MTD / action
F1 LMS01	3 h 45 min	Misses 3-hour RTO by 45 min	Within 4-hour MTD only narrowly; exam deadline already missed.
F2 FIN01	3 h 30 min	Meets 4-hour RTO	Within 8-hour MTD; reconcile provider records.
F3 REG01	4 h	Meets 6-hour RTO	Within 10-hour MTD; preserve application timestamps.
F4 HR01	4 h 30 min	Meets 12-hour RTO	Within 20-hour MTD; verify bank submission.
F5 FS01	2 h 30 min	Meets 24-hour RTO	Within 48-hour MTD; verify integrity and permissions.

DB01's latest backup is from 07:00, so there is a possible one-hour data-loss window, and the proposed RPO is met only if the backup chain works. AUTH01's midnight backup is eight hours old, which exceeds its one-hour RPO. FS01's 23:00 Day -1 backup is nine hours old, missing the eight-hour RPO by one hour. These gaps need careful reconciliation and honest reporting.

5.2 Tabletop exercise (45 minutes)

Time / inject	Decision and revised action	Owner / rationale
0–10 minutes: start	Confirm roles, contact tree, site safety and incident log.	BC/DR Coordinator; establish decision authority.
Minute 10: closure extends to five days	Extend alternate/remote arrangements, staffing, supplies, learner communications and cost approval.	Director / Facilities; 72-hour assumption no longer holds.
Minute 20: only 15 workstations	Do not exceed capacity. Use controlled sittings or approved rescheduling for 40 centre-dependent learners.	Academic Director; protect safety and assessment integrity.
Minute 30: DB01 increment fails integrity; unusual backup admin logon	Stop suspect chain, preserve logs, restrict affected credentials, involve Incident Response, check immutable/offline copy and choose independently verified point.	IT Lead / Incident Response; integrity failure plus privileged anomaly may indicate compromise.
Minute 40: payments absent from restored finance records	Pause final confirmation for affected transactions, obtain provider report, record references, dual-review and reconcile before posting.	Finance Manager; avoid duplicate or incorrect ledger entries.

5.3 Corrective action register

Priority / action	Owner	Target	Evidence of completion

P1 Separate backup and production administration; enable MFA	IT Lead	30 days	Separate accounts, MFA report and signed access review.
P1 Implement immutable/offline backup and confirm provider access independence	IT / Procurement	60 days	Configuration evidence and successful independent restore.
P1 Test full DB01 chain and one transaction end-to-end	DR Lead	30 days	Signed test log, duration, checkpoint and exceptions.
P1 Approve assessment continuity for 40 learners and 25 seats	Academic Director	14 days	Approved procedure and exercise record.
P2 Improve AUTH01 checkpoint frequency	IT Lead	60 days	Hourly backup evidence and successful test.
P2 Document provider reconciliation process	Finance Manager	30 days	Matched transaction sample, exception log and dual approval.
P2 Run five-day outage exercise	BC/DR Coordinator	90 days	Exercise report, assigned owners and closed actions.

TASK 6: GOVERNANCE AND PERFORMANCE MEASURES

6.1 Ownership and review

The Institute Director is responsible for and approves the BCDR plan. The BC/DR Coordinator keeps it updated, the IT Lead manages technical recovery steps, and each function owner approves minimum service and reconciliation controls. The plan should be version-controlled, limited to authorized staff, and stored in an approved central place, with an offline emergency copy. testing calendar should include quarterly restore tests for critical data, semi-annual technical recovery testing, an annual full continuity exercise and tabletop exercises after major changes. Review the plan after a serious incident, failed restoration, supplier or site change, significant application change, material audit finding or change in business requirements. Recovery-role staff should receive role-based training and annual refreshers.

This is in accordance with NIST SP 800-34 Rev. 1 and ISO 22301:2019, not a claim of certification. NIST SP 800-34 covers contingency planning, BIA, preventive controls, recovery strategy, plan development, testing/training and maintenance (Swanson et al., 2010). ISO 22301 delivers a framework for establishing, implementing, maintaining and continually improving business continuity arrangements (ISO, 2019).

Assignment output	NIST SP 800-34 alignment	ISO 22301 theme
BIA and impact calculations	Conduct BIA and prioritise systems	Impact analysis and continuity requirements
RTO/RPO/MTD and backup design	Set recovery requirements and preventive controls	Continuity strategy and resources

Package comparison and budget	Select recovery strategy	Planning and resource provision
BC/DR plan, RACI and runbook	Develop contingency procedures	Operational planning, roles and communication
Baseline schedule and tabletop	Test, train and exercise	Performance evaluation and exercises
KPIs, action register and calendar	Maintain plan	Monitoring and continual improvement

Flood and climate risks should guide site selection and continuity planning. The institute should check flood exposure, drainage, access routes, power backup, equipment elevation, alternate site location, and supplier dependencies.

KPI / method	Source	Target / frequency	Accountable owner
RTO achievement = functions restored within RTO ÷ functions tested × 100	DR logs and business acceptance	≥90% by month 6; 100% for F1–F3 by month 12; quarterly	IT Lead
Restore-test coverage = critical systems restored successfully ÷ systems scheduled × 100	Backup platform and test reports	100% of critical systems quarterly	IT Lead
Corrective-action closure = actions closed on time ÷ actions due × 100	Action register	≥90% quarterly; overdue P1 escalated monthly	BC/DR Coordinator
Training completion = recovery staff trained ÷ recovery staff assigned × 100	Training register	100% annually; new role holders within 30 days	HR / BC/DR Coordinator
Backup integrity success = restore points passing checks ÷ restore points tested × 100	Backup validation logs	100% for selected recovery points; failures escalated immediately	IT Lead
Reconciliation completion = outage transactions reconciled and approved ÷ transactions logged × 100	Temporary registers and finance/admissions logs	100% before incident closure; after each outage	Function owner

6.3 Maturity assessment and target

Using a five-level scale (1 Initial, 2 Repeatable, 3 Defined, 4 Managed and measured, 5 Optimised), I rate Northbridge at Level 2 (Repeatable) for this exercise. Some backups and a warm-site drill are in place, but backup access independence is not documented, AUTH01 and FS01 checkpoints do not meet the proposed RPOs, and the assessment schedule misses the 11:00 deadline. Activities are present, but they are not always controlled or proven to meet business needs.

The goal for the next 12 months is Level 3 (Defined), with some Level 4 measures for key recovery tests. The institute should have approved, version-controlled procedures; named owners; independently protected backups; tested recovery goals; an annual exercise; tracked corrective actions; and KPI reporting to leadership.

CONCLUSION

The assessment shows that Northbridge needs to handle the disruption as a business event, not simply a technical recovery. Teaching and assessment have the most urgent deadline, but shared infrastructure controls when LMS01 can be restored. The assessment service returns at 11:45, which is too late for the 11:00 exam. It therefore becomes necessary to have a written temporary assessment procedure.

The warm package, which includes independent protected backups and training/testing, costs NGN 2,100,000 per year and is the recommended affordable choice, provided leadership accepts that it does not guarantee the proposed targets. Significant upgrades include independent backup management, shorter checkpoint gaps, tested restore chains, controlled temporary records, and regular exercises. The team should declare this only after recovery, technical checks, and business approval.

People and premises are part of the recovery requirement, not secondary considerations. The Institute must confirm that the alternate facility is safe, has accessible entry and toilets, has adequate power and connectivity, and can support invigilation without exposing learner information to other users.

Supplier dependency also needs active management. Microsoft 365 email and identity remain available, but do not assume mobile internet is unlimited or stable. The continuity coordinator should identify at least one alternative approved communication route and keep current contact details for the payment provider, bank, backup provider and alternate-site contact.

Information requirements include the candidate list, examination content, approved payroll file, payment-provider transaction reports, admission timestamps and access-controlled records. The team should identify the authoritative copy for each record and prohibit uncontrolled copies on personal devices. Where temporary registers are necessary, each entry should have a unique reference, timestamp, preparer, approver and later reconciliation status. This reduces the risk of duplicate processing and gives the Institute an audit trail after the main systems return.

Additional recovery objective and risk considerations

The proposed plans reflect business deadlines hence, the Institute should validate them through a workshop with each function owner and document the assumptions behind them. For example, a three-hour RTO for F1 is meaningful only if the assessment can be delivered fairly, learners can authenticate, the question set is available, and results can be recorded safely. A technical service that is online but cannot support a valid assessment should not be counted as recovered.

The RPO gaps require management decisions because the current checkpoints do not all meet the proposed targets. If DB01's incremental chain fails, the latest 07:00 point cannot be used simply because the catalogue lists it. The team must find the most recent independently verified point, calculate the resulting data gap and communicate the consequences to the business owner. Finance, admissions and payroll must then use source evidence to rebuild missing transactions. Where evidence is insufficient, the entry should remain an exception for investigation rather than being guessed or silently recreated.

I recommend we treat the warm option as a selected strategy, not a guarantee. Before approving it, management should request evidence of what the quoted 3–5-hour window includes, whether it assumes that the network, directory and database backups are usable, whether the supplier has reserved capacity, and how the service is tested. Contractual recovery commitments should identify measurement points, exclusions, escalation routes and the evidence provided after an exercise, then verify a restitution claim through a witnessed test using the Institute's own acceptance criteria.

The NGN 900,000 difference between the warm package total and the annual limit is not automatically spare money. It may be needed for additional connectivity, transport, accessible facilities, replacement equipment or unplanned recovery work. Finance should retain a contingency allowance and require approval for any spend that changes the chosen strategy. If the Institute chooses not to fund the hot option, the Director should document the remaining risk that a critical service may not return within the shortest target, particularly when the examination deadline is close.

The incident log will show the record, when the disruption was identified, who declared the plan active, decisions made, systems affected, recovery points selected, communications issued, approvals received, and pending issues. All evidence should be stored in a controlled location with appropriate access restrictions. This log supports after-action review and helps management determine whether a missed target was caused by technology, a dependency, an unclear decision right, supplier performance, or an unrealistic recovery assumption.

Do not close corrective actions merely because a task has been labeled complete. Each action needs evidence that the control works. For example, separating backup administrator accounts should be followed by an access test demonstrating that a production administrator cannot delete immutable backups. Improving backup frequency should be followed by a restore test that measures the actual data-loss window. Measure training by attendance and participants' ability to perform assigned actions during an exercise, not attendance alone.

The maturity assessment is based only on the fictional evidence supplied in the scenario. A Level 2 assessment is reasonable because some recovery functions and a drill exist, but important elements remain unverified. Progress to Level 3 requires documented procedures, named owners, approved recovery objectives, repeatable testing and evidence that corrective actions are managed. Leadership should review the dashboard quarterly and require explanations for missed targets and ensure the plan works.

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